

Every Trade Provable. Every Loss Capped.

An autonomous 0DTE options agent on Alpaca. Max loss capped by construction, and every decision it makes is journaled.

LIVE ALPACA PAPER · 3 SESSIONS

+79.1%

\$100,000 → \$179,087

MAX LOSS

= premium paid

Single-leg long only

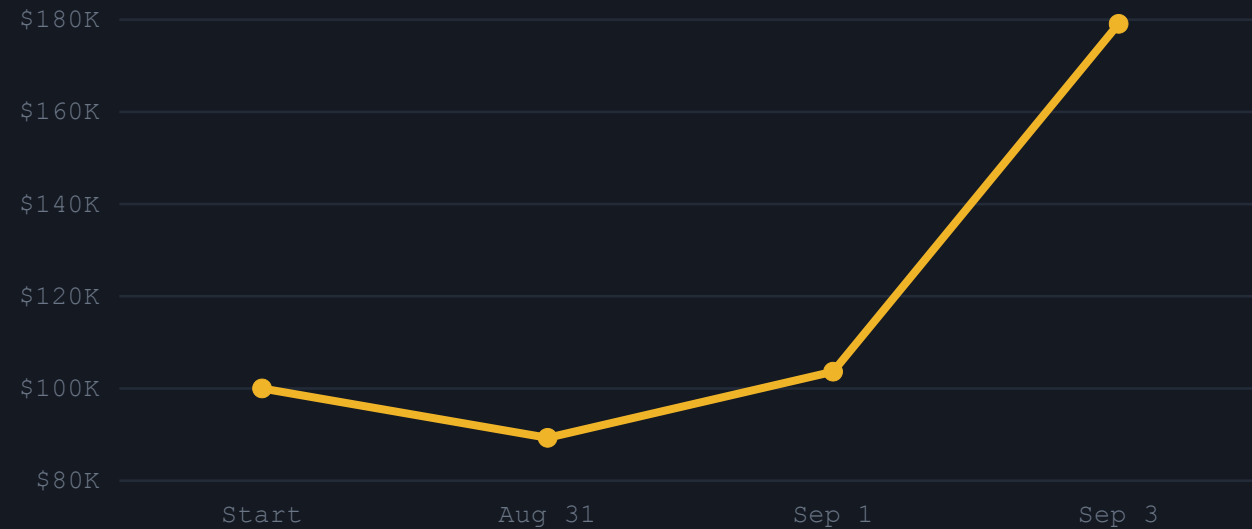
PAPER ACCOUNT

PA38HG4D9653

Opened fresh for this event at \$100,000

A fresh \$100,000 account, 3 sessions, +79.1%

EQUITY · ALPACA PAPER PA38HG4D9653



REALISED BY SESSION

Aug 31

-\$10,716

19 in / 16 out

Sep 1

+\$14,353

5 in / 9 out

Sep 3

+\$75,450

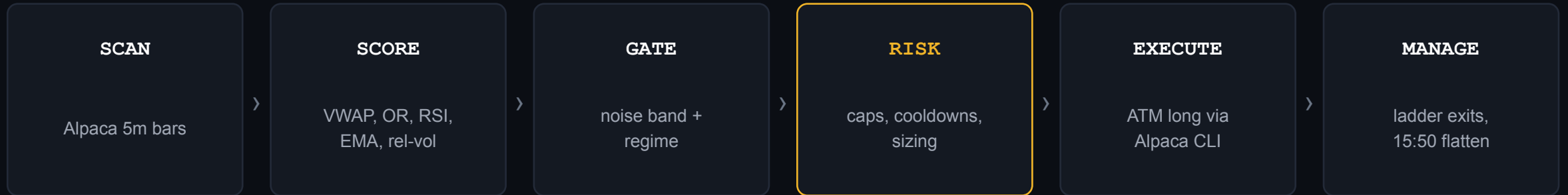
10 in / 18 out

NET

+79.1% · \$179,087

34 entries and 43 exits on SPY / QQQ / IWM, every one placed through the Alpaca CLI and booked at the real broker fill. Positions are flat at the close of every session - the agent never carries a ODTE into expiry.

One loop, six stages, no human in it



DEFINED BY CONSTRUCTION

**Max loss
= premium**

Single-leg long only. The ceiling is set by the instrument, not by code that has to remember.

VALIDATED, NOT GUESSED

264 sessions

of real OPRA option bars and 4,544 alerts stand behind every control that ships.

FULLY AUTONOMOUS

**Docker,
unattended**

One session per weekday in an isolated container. Nobody approves a trade.

Three independent tests. A trade needs all three.

SCORED SIGNAL

VWAP, OR break, RSI,
EMA, relative volume

NOISE BAND

is the break bigger than
the symbol's own noise?

LEADER REGIME

5 of 8 mega-caps above
their 20-day average

ORDER

ATM long via the
Alpaca CLI

THE HALF-OPENING-RANGE BAND

noise, not signal

A break only counts if it clears half of that symbol's own opening range. The bar is set by the morning's actual behaviour rather than a fixed number, so a quiet SPY and a wild IWM are not judged the same way.

LEADER BREADTH

removes, never adds

5 of 8 mega-caps above their 20-day average as of the last completed session. The regime can only rule out the opposed direction - it never creates a trade the score did not already propose.

A trade needs the score at or above 0.70, price outside the band, and the direction agreeing with the regime. Any one of the three says no and nothing is placed.

Risk is structural before it is procedural

Single-leg long options only. Maximum loss is the premium paid - enforced by the instrument, not by code that has to remember.

ONE-LOT GUARD

Never a second open lot on a (symbol, direction) already held.

ONE DIRECTION / UNDERLYING

No simultaneous call and put on the same symbol.

ENTRY DEDUP · 30 MIN

Entry-anchored, so a persistent score cannot stack the same name.

ADAPTIVE SIZING

Trimmed to the account's real options buying power - an expensive contract sizes down, never rejects.

SESSION WINDOW

No entry after 15:00 ET. Hard flatten at 15:50 - a 0DTE never reaches expiry.

ORPHAN SWEEP

Any option position the agent did not open is flattened at startup and at EOD.

Exits ladder in both directions: +40% scales out half and trails the runner by 40% of its peak gain · -20% sells half, -40% the rest · a 30-minute stop closes anything still under water · 15:50 flatten.

Every order goes through the Alpaca CLI

ORDERS PLACED THROUGH THE CLI

The agent runs `alpaca order submit` and waits for the order to fill. It records the price it actually got, so the profit and loss is real money in and out - not an estimate from a quote.

PAPER ACCOUNT, CHECKED EVERY TIME

Before it places anything, the agent confirms `alpaca doctor` is pointing at the paper endpoint. If it is not, the agent does not trade.

NEVER A DUPLICATE ORDER

If a submit times out, the agent looks the order up by its own ID before trying again. The result is one order, or none - never two.

RUNS ON ITS OWN, AND LOGS IT

Prices and contracts come from Alpaca's market data API. It runs in its own container, once each weekday, and writes a log line for every decision.

Most agents show the trades they took

This one shows the ones it refused.

```
logs/decisions-2026-09-03.jsonl
```

```
{ "symbol": "QQQ",  
  "score": 0.68,  
  "would_have_direction": "put",  
  "noise_band": { "state": "inside" },  
  "gate": { "go": false,  
    "rationale": "score 0.68 < min 0.70" },  
  "order": null }
```

ONE RECORD PER SCAN

Full reasoning

Score, every signal, band state, gate verdict, risk verdict, order.

WHY IT MATTERS

Nothing is implicit

You can reconstruct every decision the agent made, and every one it declined to make.

logs/ in the repo carries 14 journal records across 2 session files - clone it and check any decision against the record.

AI logic · Risk gates · Alpaca infrastructure

Alpaca paper PA38HG4D9653 · single-leg long 0DTE on SPY / QQQ / IWM · **\$100,000 → \$179,087 (+79.1%) across 3 sessions, 34 entries / 43 exits**

1 · AI logic

- Scorer reduces Alpaca 5m bars to one number: VWAP position, opening-range break, RSI, EMA stack, relative volume - plus a proposed direction.
- Half-opening-range noise band: is the break larger than the symbol's own morning noise?
- Leader breadth: 5 of 8 mega-caps above their 20-day average as of the last completed session. Only ever removes the opposed side.
- A Claude layer argues bull and bear over the scored facts and returns strict JSON - and can only turn a go into a no-go.
- It can never resurrect a rejected trade, so the deterministic path is a strict subset of the AI path.
- A missing key or unparseable reply degrades to the deterministic gate and says so in the journal.

2 · Risk gates

- Single-leg long only: max loss is the premium paid, set by the instrument.
- One open lot per (symbol, direction); one direction per underlying.
- 30-minute entry-anchored dedup, so a persistent score cannot stack the same name.
- Sizing trimmed to the account's real options buying power - expensive contracts size down, never reject.
- No entry after 15:00 ET; hard flatten at 15:50, so a 0DTE never reaches expiry.
- Orphan sweep at startup and EOD for positions the agent did not open.
- Exits: +40% sells half and trails the runner; -20% half, -40% the rest; 30-min stop if under water.
- 174 network-free tests cover all of it.

3 · Alpaca infrastructure

- Every order through the Alpaca CLI, not the SDK: ``alpaca order submit``, then poll to a terminal state.
- So the agent books the real broker fill, not a quote - cash P&L is buy-fill against sell-fill.
- Refuses to place anything unless ``alpaca doctor`` resolves `paper-api.alpaca.markets`.
- An ambiguous submit is reconciled by `--client-order-id` before it will ever resubmit.
- Market data - bars, contract discovery, ATM quotes - via the Alpaca Market Data API.
- Runs unattended in an isolated Docker container, one session per weekday.
- Every scan appends one JSONL record with the full reasoning to logs/.